

Highlights

Dose-Response Curve Alignment via Synthetic Data for Physiologically Conformant Glucose Forecasting in Type 1 Diabetes

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- DOCAS aligns a model's interventional audit curve—not its feature attributions—to a ReplayBG-derived insulin target via synthetic augmentation.
- Observational CGM data confound insulin with hyperglycaemia, corrupting counterfactual dose–response behavior.
- OhioT1DM (12 participants): DOCAS beats baseline, monotone LightGBM, and Prendin np-/p-LSTM on audit-curve alignment with negligible RMSE cost.
- Open-source Python package: user-defined target function, context rule, and model.
- ReplayBG on patient 588: DOCAS improves time-in-range and reduces misaligned-baseline hypoglycaemia.
- Ablation study shows the augmentation mechanism generalizes to flat, interaction, multi-feature, and full-range targets.

Dose-Response Curve Alignment via Synthetic Data for Physiologically Conformant Glucose Forecasting in Type 1 Diabetes

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ABSTRACT

Machine-learning glucose models are increasingly embedded in type 1 diabetes (T1D) decision support and closed-loop systems, yet prediction accuracy alone does not guarantee safe therapeutic use. Observational CGM data induce strong *covariate bias*: boluses are administered when glucose is already elevated and are collinear with prandial carbohydrate intake, so the observational joint distribution conflates the glucose-lowering effect of insulin with the hyperglycaemic contexts that trigger dosing. Empirical risk minimization on such data can achieve low error by fitting associative patterns—future glucose remains high *because* insulin coincides with high CGM and meals—rather than encoding how glucose would change under counterfactual bolus perturbations. Diagnosing this with feature attributions reveals the symptom—e.g., positive SHAP contributions for insulin—but does not repair the underlying interventional behaviour. We introduce *DOCAS* (dose–response curve alignment via synthetic augmentation), a lightweight, model-agnostic post-training procedure that does *not* align feature attributions. Instead, it aligns the model’s *interventional audit curve*—the response traced by sweeping bolus insulin under a *do*(·)-style query with context held fixed—to a user-specified *target function* τ on hyperglycaemic contexts. DOCAS augments observational data with synthetic samples labeled by τ where an intervention is meaningful. In this study τ is derived by replaying the insulin pharmacokinetic and glucose–insulin equations of ReplayBG; the same mechanism accepts arbitrary target functions and surfaces. On held-out OhioT1DM data from all 12 participants, DOCAS sharply reduces alignment error relative to a strong LightGBM baseline, a hard monotone-constrained LightGBM comparator, and a Prendin-style LSTM family (np-LSTM, p-LSTM), while essentially preserving held-out RMSE. In ReplayBG digital-twin simulation of corrective-bolus decision support on patient 588—the case study of Prendin et al.—DOCAS LightGBM increases time-in-range at both horizons and reduces the PH30 hypoglycaemia induced by a misaligned baseline controller. An ablation study shows the augmentation loop generalizes to flat, interaction, full-range, and two-dimensional targets. DOCAS is an open-source Python package in which practitioners supply their model, intervention feature, context rule, and target function—a more flexible repair strategy than architecture-specific or monotonicity-only alternatives.

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1. Introduction

Machine-learning glucose models are increasingly used in type 1 diabetes (T1D) decision support and public prediction benchmarks [1–6]. Model selection nevertheless remains dominated by held-out prediction error (RMSE, MAE), even though two models with indistinguishable accuracy can encode radically different—and clinically incompatible—input–output relationships [7–10]. In observational CGM data, insulin dosing is strongly correlated with concurrent or antecedent hyperglycaemia and meal intake—a form of *treatment confounding* familiar from causal inference [11]. Doses are delivered in response to glycaemic state, not assigned experimentally. Under such *covariate bias*, empirical risk minimization learns associative patterns that predict well on held-out rows yet fail under interventional use, when a decision support system (DSS) sweeps candidate bolus doses holding other inputs fixed [7, 12–14]. Prendin et al. showed that an LSTM with accuracy comparable to a physiologically preprocessed counterpart assigned *positive* SHAP contributions to insulin and forecasted glucose *rising* after bolus administration; only the correctly attributed model improved glycaemic outcomes in corrective-bolus DSS [7, 15–17]. The model can therefore appear competent on observational metrics while encoding the wrong interventional dose–response map.

Architectural priors—meal and insulin absorption filters before recurrent layers [7, 18–20]—can partially decouple confounded inputs, but require task-specific design and do not guarantee correct audit-curve shape on hyperglycaemic contexts. Kushner et al. showed that neural predictors trained for accuracy alone routinely violate monotonicity constraints and proposed conformance verification to detect such failures [13, 21].

We address this gap with *DOCAS* (dose–response curve alignment via synthetic augmentation), an open-source Python package for observational models whose interventional behaviour must match domain priors. DOCAS aligns the model’s *interventional audit curve*—the trajectory $u \mapsto \hat{g}(x, u)$ obtained by sweeping the intervened feature under a *do*(·)-style query with context x held fixed—to a user-specified *target function* $\tau(x, a)$, not to a *feature attribution* (a scalar such as a SHAP value for one prediction). The audit curve is exactly the counterfactual a bolus-sizing DSS reads off when choosing a dose [7, 12, 22]; τ is the domain prior for how the forecast *should* change under that probe. When we write *dose–response curve* without a qualifier, we mean this generic dose-to-outcome relation and then specify *audit* (model) or *target* (prior). Rather than modifying model architecture or applying post-hoc explanation alone, DOCAS *trains* a user-chosen model on observational rows plus synthetic counterfactuals so that the audit curve matches τ : the practitioner specifies (i) which feature is intervened, (ii) a context predicate that selects anchor states, and (iii) τ at each anchor and dose level. In T1D we instantiate τ with a ReplayBG-derived correction-bolus response on hyperglycaemic anchors (Section 3.2), but the same loop accepts other curves, surfaces, or null constraints (Section 4.3). The procedure is model-agnostic—here demonstrated with LightGBM and LSTM [23].

Although validated on T1D, DOCAS addresses a general failure mode whenever treatment variables are confounded with outcome severity [11, 14]. Because target shape, context selection, and model family are all user-defined, the method is more flexible than architectural preprocessors, hard monotonicity constraints, or simulator-coupled training objectives that lock alignment to one network or one causal graph. Section 4 evaluates the T1D default on (i) cohort-wide OhioT1DM forecasting with dose–response alignment metrics, (ii) ReplayBG in-silico replay on patient 588, and (iii) an ablation study (Section 4.3) that stress-tests alternate targets. The open-source package exposes the same interface for other domains (Section 3.3). Section 2 contrasts DOCAS with attribution-based alignment (including causal, do-operator attributions) and with interventional dose–response estimation (Table 1). Section 5 discusses implications for explainable AI and transfer beyond glucose forecasting.

2. Related Work

Prendin et al. expose the central failure mode: two LSTM glucose models with similar RMSE produce opposite corrective-bolus decisions because the standard model assigns positive SHAP contributions to insulin, whereas a physiologically structured p-LSTM recovers the expected insulin-lowering and carbohydrate-raising signs [7, 15]. Related glucose-XAI studies use attributions diagnostically [17, 24]. These works align, or check, a *feature attribution*—a scalar per-feature importance summarising one prediction. DOCAS instead aligns the model’s *interventional audit curve*, the full trajectory $u \mapsto \hat{g}(x, u)$ obtained by intervening on the bolus feature while holding the hyperglycaemic context fixed. An attribution can carry the correct insulin sign yet still hide a curve that is flat, non-monotone, or wrongly shaped over the dose grid a DSS actually sweeps; the audit curve makes that shape the object of alignment. Moreover, SHAP under dependent features is not a causal effect, so a correct sign is neither necessary nor sufficient for a correct counterfactual response; physiological knowledge or interventions are still needed to define the desired curve [7, 13, 14, 25].

The closest general literature is *explanation-guided learning*, which feeds feature attributions back into optimization. Right for the Right Reasons penalizes input gradients on features experts mark as irrelevant [26]; HINT uses a ranking loss to match gradient-based importance to human attention [27]; and CDEP directly penalizes contextual decomposition attributions that conflict with prior knowledge [28]. Erion et al. generalize this idea with Expected Gradients and attribution priors over properties such as sparsity, smoothness, and biological structure [29]. These studies establish that attributions can be differentiable training signals, but they mostly align feature masks, rankings, or global priors in neural classifiers; they do not specify a numeric treatment dose–response curve or demonstrate T1D decision support.

A distinct line of work makes attributions themselves causal by evaluating them under Pearl’s *do*-operator rather than observational conditioning [22]. Janzing et al. argue that dropping a feature in a Shapley computation should mean intervening on it, $\mathbb{E}[Y \mid do(X_S = x_S)]$, not conditioning on it [30]; do-Shapley and causal Shapley values build on this to attribute a model’s output to interventions on individual features [31, 32]. This “do-SHAP” family shares our premise—that observational attributions conflate association with intervention—but differs from DOCAS in three ways. First, its output is still an *attribution*: a scalar causal contribution per feature for one prediction, not the full response curve $u \mapsto \hat{g}(x, u)$ a DSS sweeps. Second, it is a read-out that *measures* an already-trained model rather than a training procedure that *repairs* it. Third, it needs a specified causal graph or SCM and identifiability of the interventional estimand from data. DOCAS requires neither a causal graph nor identifiability: because the insulin dose–response effect is not identified from observational T1D data, we do not estimate $\mathbb{E}[Y \mid do(u)]$ but instead *impose* a physiological target on the audit curve and fit the model to it. In short, do-SHAP audits an interventional attribution; DOCAS aligns the interventional curve.

A parallel literature evaluates attributions against known references. XAI-BENCH uses synthetic data to compute ground-truth Shapley values [33], while OpenXAI scores feature, sign, and rank agreement with ground-truth explanations alongside faithfulness and stability [34]. This motivates explicit alignment scores, but the reference differs from ours: those benchmarks test whether an explainer recovers a known model or data-generating function, whereas DOCAS scores a model’s audit curve against a stated target function τ . Because counterfactual responses are not identified from observational T1D data alone, τ is a domain prior rather than observed ground truth.

A further body of work aligns models to interventions by *estimating* a dose–response curve from observational data. Continuous-treatment causal methods de-confound treatment from covariates—balancing representations, multi-head dose architectures such as DRNet, or amortised interventional predictors—to recover $\mathbb{E}[Y \mid do(t)]$ under strong ignorability and positivity [11, 35]. DOCAS and these methods both concern interventional dose–response curves, but from opposite ends: they *recover* an identifiable estimand from data, whereas DOCAS assumes the effect is *not* identifiable (insulin is dosed precisely in response to hyperglycaemia, violating ignorability) and therefore substitutes a stated target function τ for the missing experiment. DOCAS also aligns the working forecaster the DSS already uses via data augmentation, rather than fitting a separate causal-effect estimator with its own architecture.

Within glucose forecasting, alignment has instead been imposed through physiological preprocessing [7, 18, 19], formal conformance penalties [13, 21], and hard monotonic models [36, 37]. LightGBM provides the latter through monotone constraints [23], which we include as a comparator in Table 2. Other approaches encode mechanistic inputs [38], align internal representations to a simulator by Interchange Intervention Training [39], or constrain a synthetic CGM generator [40]. Outside T1D, expert-gated synthetic dose–outcome supervision has also been proposed for personalized drug dosing [41]. Table 1 summarizes the main distinctions.

DOCAS is therefore an explanation-guided repair method in spirit, but it aligns the interventional audit curve to a target function τ rather than any attribution: it does not optimize SHAP values, causal or otherwise. Synthetic rows train any model against a full user-defined target over the intervention grid, not only an attribution sign, feature mask, or monotonic split constraint, and without assuming the effect is identifiable from data. The ReplayBG experiments (Section 4.2) then test whether this aligned counterfactual behaviour improves DSS safety.

3. Methods

All code to reproduce the results reported in this manuscript is available on GitHub ([GITHUB_REPOSITORY_URL]).

Figure 1 summarizes the DOCAS pipeline: a user-supplied model is trained on observational data (Section 3.1), anchor contexts are selected by a context rule and swept along a user-defined target function (Section 3.2), a new model is trained on real and synthetic rows (Section 3.2), and the result is scored on held-out alignment and RMSE (Section 3.2, Section 3.3). The subsections below instantiate this interface for T1D glucose forecasting; Section 4.2 embeds the aligned model in decision support.

Table 1

Strategies for aligning models to the physiologically expected insulin response (glucose should fall, monotonically, as corrective insulin increases), including general attribution- and intervention-based baselines. *Aligns curve*: targets the full interventional audit curve $u \mapsto \hat{g}(x, u)$ rather than a scalar feature attribution. *Repairs model*: changes training or architecture rather than only measuring, verifying, or generating data. *Model-agnostic*: applicable without a task-specific network redesign. *T1D validation*: evaluated on glucose forecasting or insulin decision support with a concrete audit-curve or attribution check.

Strategy	Representative studies	What is aligned	Aligns curve	Repairs model	Model-agnostic	T1D validation
Attribution alignment / diagnosis	[7, 17, 26, 28, 29]	Feature attributions (SHAP sign, gradients) matched to expert priors; scalar per prediction	×	Partial	Partial	Partial
Causal / do-operator attributions (do-SHAP)	[30–32]	Per-feature causal contribution $\mathbb{E}[Y \mid do(x_S)]$; measures a fitted model, needs a causal graph	×	×	✓	×
Interventional dose-response estimation	[11, 35]	Estimates $\mathbb{E}[Y \mid do(t)]$ from data via balanced representations / dose heads under ignorability	✓	✓	×	×
Physiological preprocessing / architecture	[7, 18–20]	Input filters (IOB/COB) or a p-LSTM layer so SHAP attributions match insulin/CHO PK	×	✓	×	✓
Conformance verification + penalty training	[13, 21]	Range-propagation checks and training penalties for insulin monotonicity in NN glucose predictors	Partial	✓	Partial	✓
Hard monotonic architecture	[36, 37]	Architecturally monotone input-output maps (lattice networks, constrained-activation MLPs)	Partial	✓	Partial	×
Hard monotonic constraint, gradient-boosted trees (this paper, Table 2)	[23]	LightGBM monotone constraints on the insulin split direction	Partial	✓	✓	✓
Mechanistic feature engineering, LightGBM	[38]	Optimized Bézier macronutrient dynamics (optionally from meal images via mLLM) as LightGBM inputs	×	✓	✓	✓
Monotonicity-constrained synthetic-data generator	[40]	GAN generator constrained so BG decreases with insulin, increases with CHO	Partial	×	×	✓(sim.)
Synthetic dose-response supervision (DOCAS, ours)	—	User-defined target function τ on anchors selected by a context rule; imposed on the interventional audit curve, no identifiability assumed	✓	✓	✓	✓

3.1. Models and OhioT1DM protocol

For each participant we train forecasters f_b that map a feature vector x_t (recent CGM, insulin, carbohydrate, and time-of-day channels, each min–max scaled to $[0, 1]$) to the change in CGM over a prediction horizon h . The predicted future glucose is $\hat{g}_b(x) = \text{CGM}_t + f_b(x)$, evaluated at $h = 30$ and $h = 60$ min. The primary backbone is a gradient-boosted regression tree (LightGBM [23, 42]). As comparators we also train (i) LightGBM with a hard monotone-decreasing constraint on the insulin feature [23], and (ii) a Prendin-style LSTM family matching interpretable-DL4BGP [7]: np-LSTM (raw bolus/meal impulses; LSTM(64)+Dense, 100 epochs, batch size 256), p-LSTM (offline IOB/COB smoothing plus Dense with tanh activation), and DOCAS LSTM (the same synthetic-augmentation loop applied to the p-LSTM backbone).

Following the OhioT1DM preprocessing of Marling et al. [3] and Prendin et al. [7], each participant’s first six monitoring weeks form the training set and the final ten days the held-out test set; CGM gaps under 30 min in training are linearly interpolated and test-set gaps are left missing. Inputs are min–max scaled to $[0, 1]$ using training-set statistics. For LightGBM (and the offline features fed to p-LSTM / DOCAS LSTM), insulin and carbohydrate channels are converted to insulin-on-board and carbohydrate-on-board traces (rapid bolus IOB plus active basal, and meal absorption); np-LSTM instead receives the raw bolus and meal impulses. The regression target is the *change* in CGM over the horizon (ΔCGM), not absolute future glucose; held-out RMSE is therefore computed on ΔCGM in mg/dL after inverse scaling. We report cohort means $\pm$ standard deviations across all 12 OhioT1DM participants. These choices

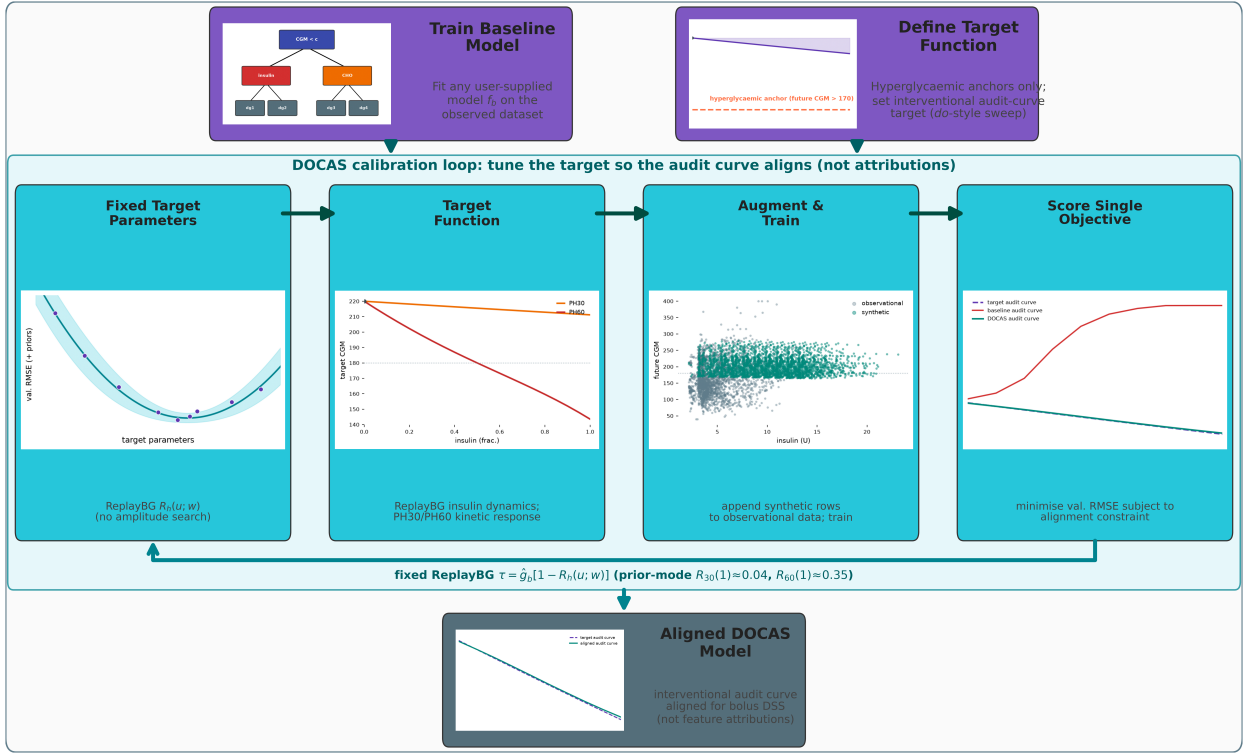


Figure 1: DOCAS pipeline (generic interface). A user-supplied baseline model is probed on anchor contexts selected by a context rule; sweeping the intervention feature traces the *interventional audit curve* $u \mapsto \hat{g}(x, u)$ —not a feature attribution—and synthetic labels on each anchor follow a user-defined target τ for that curve. The T1D preset holds the ReplayBG-derived PH30/PH60 response R_h fixed (Eqs. 1–2) and augments observational training with synthetic (x, u, τ) rows on hyperglycaemic anchors before refitting (Section 3.2).

differ from Prendin et al., who evaluated a single compliant subject (patient 588), predicted absolute future CGM, and scaled features to $[-1, 1]$. Pointwise RMSE values are thus *not* directly comparable across the two studies; we cite Prendin et al. for the attribution–DSS mismatch and replay on patient 588 (Section 4.2), not for numerical parity of forecasting error.

3.2. DOCAS dose–response alignment

DOCAS operates on two curves at each anchor context x , selected by a *context rule* (here baseline-predicted future glucose above 170 mg/dL). A *counterfactual probe* fixes x , sweeps the intervention feature a (here bolus insulin u) on a dose grid, and reads off model predictions—Pearl’s *do*(a) query with non-intervened features held at their values in x [22]. The *interventional audit curve* $u \mapsto \hat{g}_b(x, u)$ is that trajectory: the model’s dose–response map at x , not a scalar feature attribution. The *target function* $\tau(x, u)$ (Eq. 2) is the domain prior for how the forecast *should* change; synthetic rows use τ as their label. *Alignment error* is the RMSE (mg/dL) between audit curve and target function on absolute future-CGM amplitude on held-out hyperglycaemic test contexts (Section 4.1).

We probe each trained model with an additional correction dose: hold the anchor’s observed insulin-on-board (IOB) and all remaining features fixed, and expose a separate added-dose feature $U_{\max}u$ for $u \in [0, 1]$, with $U_{\max} = 10$ U (observational rows keep this channel at zero). The resulting audit curve is what a bolus-sizing DSS reads off. In principle it should slope downward; in practice it often does not, because insulin coincides with hyperglycaemia in observational data and the learned audit curve is frequently flat, noisy, or even rising [7, 43, 44]. Because this effect is not identified from confounded observational data, we do not estimate it; we specify $\tau(x, u)$ and train the model so its audit curve matches it.

The fix is to teach the model a mechanistic target curve. On hyperglycaemic contexts—those whose baseline-predicted future glucose exceeds 170 mg/dL, just above the 140–180 mg/dL time-in-range ceiling [45]—we derive $\tau(x, u)$ from

ReplayBG's single-meal T1D model [12]. In implementation units, its insulin and glucose states obey

$$\begin{aligned} \dot{I}_{sc1} &= -k_d I_{sc1} + I(t - \gamma), & \dot{I}_{sc2} &= k_d I_{sc1} - k_{a2} I_{sc2}, \\ \dot{I}_p &= k_{a2} I_{sc2} - k_e I_p, & \dot{X} &= -p_2 X + p_2 \frac{SI}{V_I} (I_p - I_{pb}), \\ \dot{G} &= -[S_G + \rho(G)X]G + S_G G_b + Ra/V_G, & \dot{IG} &= (G - IG)/\alpha, \end{aligned} \quad (1)$$

where the observable IG is interstitial glucose and $\gamma = 8$ min is the subcutaneous-input delay. We reproduce ReplayBG's one-minute backward-Euler updates exactly. A dose $U_{\max}u$ is delivered as the same one-minute $1000U_{\max}u/w$ mU/kg correction pulse used by the replay controller for body weight w ; all states start at basal steady state with $G = IG = G_{\text{ref}} = 220$ mg/dL.

Let $IG_h(u; w)$ be the resulting interstitial glucose at horizon h . We subtract the matched no-bolus trajectory so that the simulator supplies only the incremental insulin effect, while the fitted baseline retains each anchor's meals and glucose trend:

$$R_h(u; w) = \frac{IG_h(0; w) - IG_h(u; w)}{G_{\text{ref}}}, \quad \tau(x, u) = \hat{g}_b(x) [1 - R_h(u; w)]. \quad (2)$$

The simulator parameters are fixed before OhioT1DM fitting at the modes of the ReplayBG priors used in our replay pipeline ($S_G = 0.0174 \text{ min}^{-1}$, $p_2 = 0.11 \text{ min}^{-1}$, $SI = 1.15 \times 10^{-3}$, $k_{a2} = 0.0114 \text{ min}^{-1}$, and $k_d = 0.0204 \text{ min}^{-1}$); the remaining parameters use ReplayBG defaults. Only measured body weight varies by participant. At $w = 75$ kg and $U_{\max} = 10$ U, this recurrence gives $R_{30}(1) = 0.040$ and $R_{60}(1) = 0.347$. Thus the PH30 target falls gradually, whereas the PH60 response is much larger and becomes steeper at high dose; this behaviour is generated by the simulator rather than imposed with a hyperbolic shape (Figure 2).

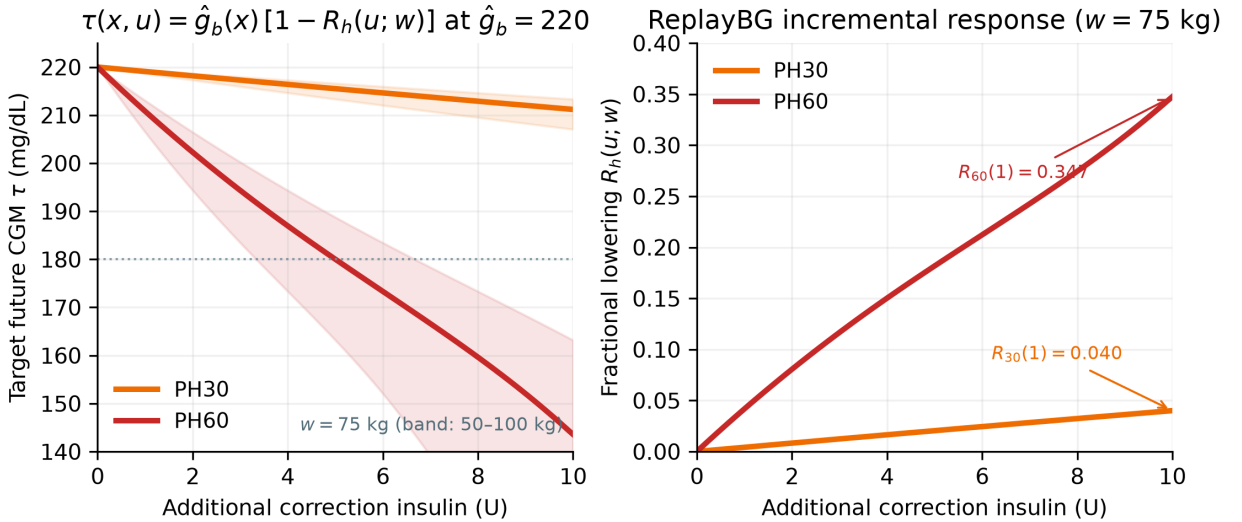


Figure 2: ReplayBG-derived DOCAS target (Section 3.2). **Left:** absolute target future CGM $\tau = \hat{g}_b[1 - R_h(u; w)]$ at $\hat{g}_b = 220$ mg/dL for PH30 and PH60; solid curves use $w = 75$ kg and shaded bands span body weights 50–100 kg. The dotted line marks 180 mg/dL. **Right:** fractional lowering $R_h(u; w)$ at $w = 75$ kg; a full 10 U correction removes about 4% of future CGM by 30 min and about 35% by 60 min. The x-axis is additional correction insulin under the counterfactual probe (separate intervention channel).

To turn this target into training signal, we form candidate (x, u) pairs on hyperglycaemic anchors with an evenly spaced interventional grid $u \in [0, 1]$, label each with (2), and append the resulting synthetic rows to the observational training set. A user-chosen model (here LightGBM or p-LSTM) is then fit on the combined real and synthetic data with a fixed synthetic-row weight, so it learns the target-curve shape on hyperglycaemic contexts while retaining accuracy on the original rows. Optional residual-correction passes reweight the synthetic labels toward the current model's residual

on the audit grid. Because R_h is taken from ReplayBG prior-mode parameters (with only body weight varying by participant), the mechanistic amplitude is not tuned against the forecasting loss: the model cannot “win” by flattening τ . Alignment in Table 2 is reported as absolute future-CGM RMSE in mg/dL.

3.3. Software interface

DOCAS is an open-source Python package that exposes the pipeline in Figure 1 through a small, composable API. The user supplies four objects: (i) a supervised model with standard `fit/predict` methods such as gradient-boosted trees or neural nets; (ii) the intervention feature to perturb when generating counterfactual rows; (iii) a *context rule* that selects anchor states where alignment should apply (in T1D, baseline-predicted future glucose above 170 mg/dL); and (iv) a *target function* $\tau(x, a)$ (Eq. 2) that maps each anchor x and intervention level a to the desired outcome label for synthetic supervision.

The core operation is the same in every use case: train a baseline on observational (x, y) pairs, sweep the intervention grid on anchors selected by the context rule, evaluate τ on each (x, a) pair, append the resulting synthetic rows, and train a new model on the combined data. *How* τ is chosen is flexible. When domain knowledge already fixes the target completely—a null curve, a prescribed interaction surface, or simulator-derived response taken literally—the user can *hardcode* τ and run a single augmentation and training pass; Section 4.3 (“hardcoded” regime) illustrates this mode, and the main OhioT1DM experiments use the same pattern with Eq. 2 held fixed. Nothing in this interface is T1D-specific: the ReplayBG-derived insulin curve is one bundled preset, and Section 4.3 demonstrates the same entry point with flat, interaction, full-range, and multi-feature targets. This paper reports the T1D preset and evaluation protocol; the package itself is domain-agnostic.

4. Results

4.1. OhioT1DM forecasting

We evaluated six forecasters on the OhioT1DM dataset [3] under the protocol in Section 3.1: baseline LightGBM, LightGBM with a hard monotone-decreasing constraint on the insulin feature [23], DOCAS LightGBM, and a Prendin-style LSTM family (np-LSTM, p-LSTM, and DOCAS LSTM) matching the Keras LSTM(64), 100-epoch, batch-256 setup of interpretable-DL4BGP [7]. Table 2 reports cohort-mean held-out test RMSE and *alignment error*: the RMSE between each model’s interventional audit curve and the ReplayBG-derived target curve (Sections 3.2 and 3.2) on hyperglycaemic test contexts. Baseline LightGBM RMSE (≈ 22.5 and 35.6 mg/dL at PH30 and PH60) is in the same ballpark as the single-subject Ohio results reported by Prendin et al. [7], but the two forecasting evaluations should not be read as a head-to-head replication because they differ in cohort scope (12 vs. one participant), prediction target (Δ CGM vs. absolute future CGM), and input representation. Despite this observational accuracy, the unaligned LightGBM model retained substantial alignment error (14.3 ± 5.5 and 29.9 ± 15.3 mg/dL). Figure 3 shows why: cohort-mean baseline audit curves are nearly flat or non-monotone across the insulin grid, consistent with the confounded insulin signatures reported by Prendin et al. and Kushner et al. [7, 13]. Hard monotone constraints cut LightGBM alignment error only partially (10.7 ± 6.6 and 21.8 ± 6.9 mg/dL) while slightly raising RMSE (22.9 ± 2.5 and 36.0 ± 4.0 mg/dL). DOCAS reduced LightGBM alignment error to 3.5 ± 1.9 and 5.7 ± 1.8 mg/dL at PH30 and PH60 while essentially preserving held-out RMSE (22.4 ± 2.8 and 35.5 ± 4.3 mg/dL). Individual participant audit curves (faint lines in Figure 3) confirm that alignment gains are cohort-wide rather than driven by a single subject. Figure 1 shows the fixed ReplayBG-derived target τ (PH30/PH60 kinetic responses) and the synthetic rows DOCAS appends to observational training data for a representative participant.

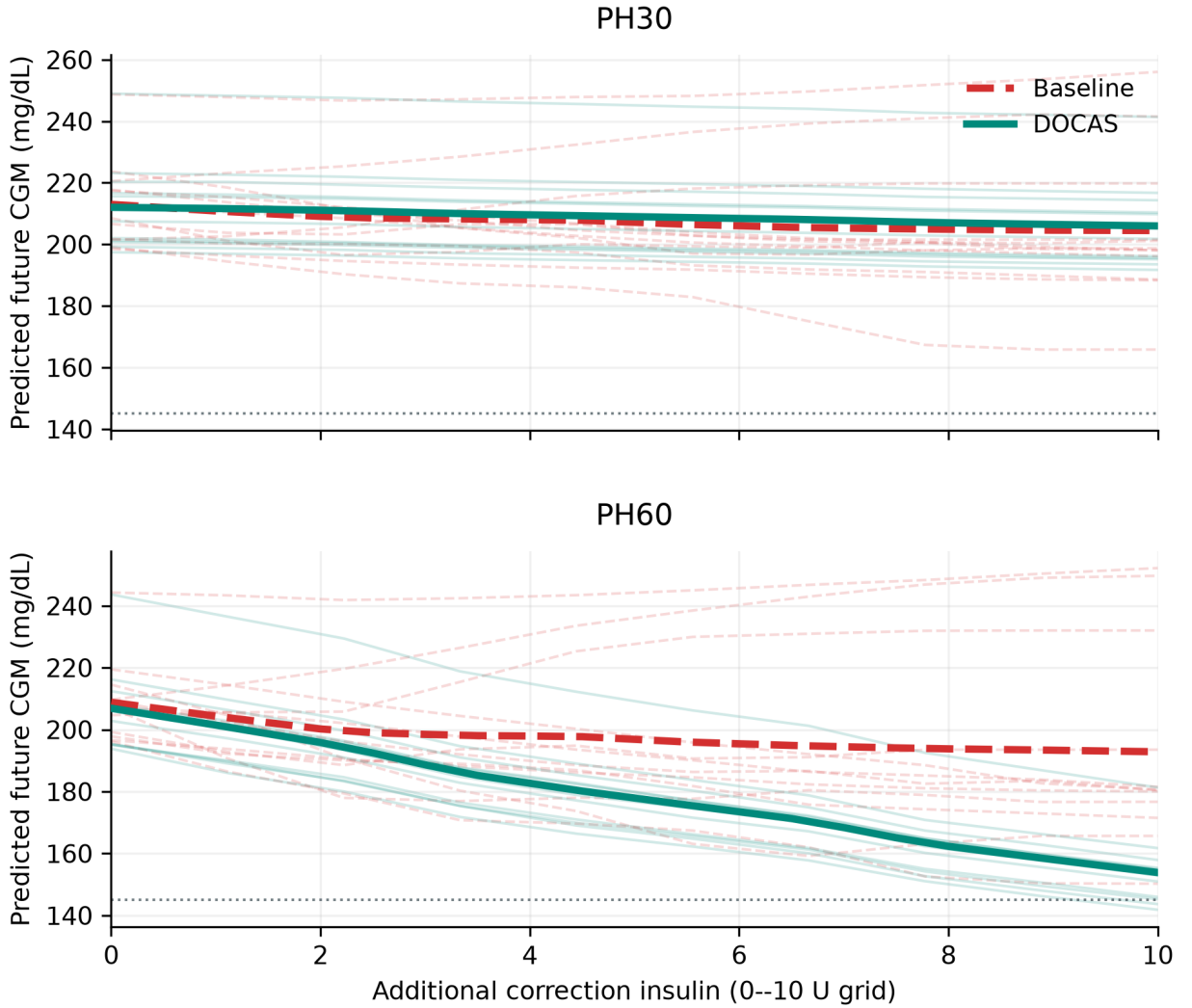


Figure 3: Held-out OhioT1DM interventional audit curves $\hat{g}(x, u)$ on hyperglycaemic test contexts (Section 3.2). **Top:** PH30; **bottom:** PH60. The x-axis is additional correction insulin on the separate intervention channel while observed IOB and other features are held fixed. Faint lines: individual participants; bold lines: cohort means. Alignment error in Table 2 is RMSE (mg/dL) between the audit curve and the target function τ in Figure 1, measured on absolute future-CGM amplitude.

Because DOCAS only requires a model with `fit/predict`, the same synthetic-target mechanism is architecture-agnostic (Section 3.3). On the Prendin LSTM family in Table 2, np-LSTM retains large alignment error (13.8 ± 9.0 and 28.5 ± 19.8 mg/dL), p-LSTM recovers much of the expected insulin-lowering shape (7.7 ± 7.8 and 20.2 ± 10.0 mg/dL), and DOCAS LSTM—the same synthetic rows on the p-LSTM backbone—tightens alignment to 1.9 ± 0.8 and 6.7 ± 4.3 mg/dL at PH30 and PH60 with only a small RMSE change. Section 4.3 stress-tests the claim on LightGBM by swapping the ReplayBG preset for flat, interaction, multi-feature, and full-range targets; the RMSE–alignment trade-off tracks the chosen τ , not a tree-specific constraint.

4.2. ReplayBG simulation

Forecasting metrics alone do not establish clinical utility when models drive therapy. We therefore embedded baseline and DOCAS LightGBM controllers in the ReplayBG digital-twin framework [12], replaying postprandial windows from patient 588—the compliant subject used by Prendin et al. [7]. For each eligible eight-hour window, ReplayBG identifies a personalized physiological model from observed insulin, carbohydrate, and CGM traces and

Table 2

Held-out OhioT1DM test RMSE and alignment error for baseline, hard monotone-constrained, and DOCAS LightGBM models, and for a second, unrelated model family (np-LSTM, p-LSTM, and DOCAS LSTM), across all 12 participants at the 30- and 60-min prediction horizons. The monotonic model applies a LightGBM monotone-constraints decrease constraint on the insulin feature [23]. np-LSTM and p-LSTM follow the architectures of Prendin et al. [7]; DOCAS LSTM applies the same synthetic-augmentation mechanism as DOCAS LightGBM to the p-LSTM architecture, demonstrating that the method is model-agnostic. Alignment error is the RMSE (mg/dL) between each model's interventional audit curve and the ReplayBG-derived target function τ on held-out hyperglycaemic test contexts (Section 3.2), measured on absolute future-CGM amplitude. RMSE is on Δ CGM (mg/dL). Values are cohort mean $\pm$ standard deviation.

Model	PH30		PH60	
	RMSE	Align.	RMSE	Align.
Baseline LightGBM	22.50 $\pm$ 2.74	14.25 $\pm$ 5.51	35.58 $\pm$ 4.26	29.93 $\pm$ 15.30
Monotonic-constrained LightGBM	22.89 $\pm$ 2.46	10.70 $\pm$ 6.64	35.99 $\pm$ 3.95	21.81 $\pm$ 6.86
DOCAS LightGBM	22.44 $\pm$ 2.75	3.52 $\pm$ 1.93	35.48 $\pm$ 4.29	5.71 $\pm$ 1.79
np-LSTM	22.38 $\pm$ 2.58	13.79 $\pm$ 8.95	35.52 $\pm$ 4.00	28.45 $\pm$ 19.76
p-LSTM	22.74 $\pm$ 2.88	7.72 $\pm$ 7.84	36.13 $\pm$ 4.45	20.17 $\pm$ 10.02
DOCAS LSTM	22.82 $\pm$ 2.89	1.85 $\pm$ 0.80	37.04 $\pm$ 4.41	6.74 $\pm$ 4.33

simulates counterfactual glucose trajectories under alternative bolus policies; windows with twin RMSE above 25 mg/dL are excluded from Table 3. The corrective-bolus DSS follows Prendin et al.: triggered two hours after meal announcement when simulated CGM exceeds 180 mg/dL, it selects the dose on a 0–10 U grid that minimizes $(\hat{g} - 120)^2 + 10u^2$ using the controller's PH-step forecast, scales the delivered bolus by a horizon gain $\sqrt{R_{30}(1)/R_h(1)}$ derived from the same ReplayBG target τ (gain 1 at PH30; ≈ 0.34 at PH60), and administers no bolus when additional insulin does not lower the predicted future glucose relative to no correction (the behaviour seen with Prendin et al.'s misaligned np-LSTM).

Without decision support (No DS), mean TIR was 49.5% with no time-below-range (TBR; Table 3)—a low mean driven by one severely hyperglycaemic eligible window. The *baseline* controller—despite similar pointwise forecasting skill—raised mean TIR to 73.2% at PH30 with 4.5 ± 0.7 U of corrective insulin, but introduced hypoglycaemia (mean TBR $3.1 \pm 6.3\%$, up to 12.5% in an individual window). Where its audit curve shows only local, non-monotone predicted dips, the Prendin grid search still issues boluses; in other misaligned contexts it would withhold CIB as in Prendin et al.'s np-LSTM [7]. At PH60 the horizon gain damps delivered doses (1.9 ± 0.5 U); baseline TBR returned to 0% while mean TIR remained 71.9%.

The *DOCAS* controller achieved the intended alignment benefit. At PH30 it raised mean TIR to 74.2% versus 73.2% for baseline, delivered slightly less corrective insulin (4.3 ± 3.5 U vs. 4.5 ± 0.7 U), and reduced mean TBR from $3.1 \pm 6.3\%$ to $2.6 \pm 5.2\%$. At PH60 its monotone audit curve selected larger raw grid optima that survive the shared horizon gain (4.0 ± 2.0 U vs. 1.9 ± 0.5 U), raising mean TIR to 75.3% versus 71.9% for baseline while keeping TBR at 0% and cutting TAR. Figure 4 dissects a representative postprandial window for patient 588 at PH30 and PH60. The upper panels show simulated CGM after the corrective-bolus decision-support (CIB) trigger (vertical dotted line, two hours post-meal): the baseline controller issues 4.0 U at PH30 and undershoots into hypoglycaemia, whereas DOCAS delivers 2.5 U and remains inside the 70–180 mg/dL band; at PH60 both arms stay in range (1.5 U baseline vs. 3.5 U DOCAS). The lower panels explain *why*: at the same decision context (simulated CGM ≈ 184 mg/dL, identical IOB and meal state), the baseline audit curve is non-monotone and predicts *rising* future glucose at higher corrective doses—yet local dips still let the CIB optimizer select a large bolus to chase the 120 mg/dL target. The DOCAS model instead predicts monotone glucose lowering across the insulin grid at both PH30 and PH60, yielding the safer PH30 bolus and the stronger PH60 correction seen in the upper panels. These results support the Prendin et al. conclusion that a correctly shaped audit curve, not RMSE alone, determines DSS safety—while showing that alignment can be achieved *post hoc* via synthetic target-curve augmentation rather than only through architectural preprocessing.

4.3. Ablation: flexibility of synthetic-target alignment

The results above use one physiologically motivated target function: the ReplayBG-derived insulin τ . In the DOCAS package (Section 3.3), that curve is merely a default *preset*; the augmentation loop itself is agnostic to target choice—it fits whichever target function or surface the user supplies on contexts matched by the context rule. We probe this

Table 3ReplayBG outcomes for patient 588 (mean $\pm$ SD).

Metric	PH30			PH60		
	None	Baseline	DOCAS	None	Baseline	DOCAS
TBR (%)	0.00 $\pm$ 0.00	3.12 $\pm$ 6.25	2.60 $\pm$ 5.21	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
TIR (%)	49.48 $\pm$ 48.62	73.18 $\pm$ 26.23	74.22 $\pm$ 25.66	49.48 $\pm$ 48.62	71.88 $\pm$ 28.75	75.26 $\pm$ 24.51
TAR (%)	50.52 $\pm$ 48.62	23.70 $\pm$ 24.39	23.18 $\pm$ 21.27	50.52 $\pm$ 48.62	28.12 $\pm$ 28.75	24.74 $\pm$ 24.51
Insulin (U)	0.00 $\pm$ 0.00	4.50 $\pm$ 0.71	4.25 $\pm$ 3.50	0.00 $\pm$ 0.00	1.88 $\pm$ 0.48	4.00 $\pm$ 2.00

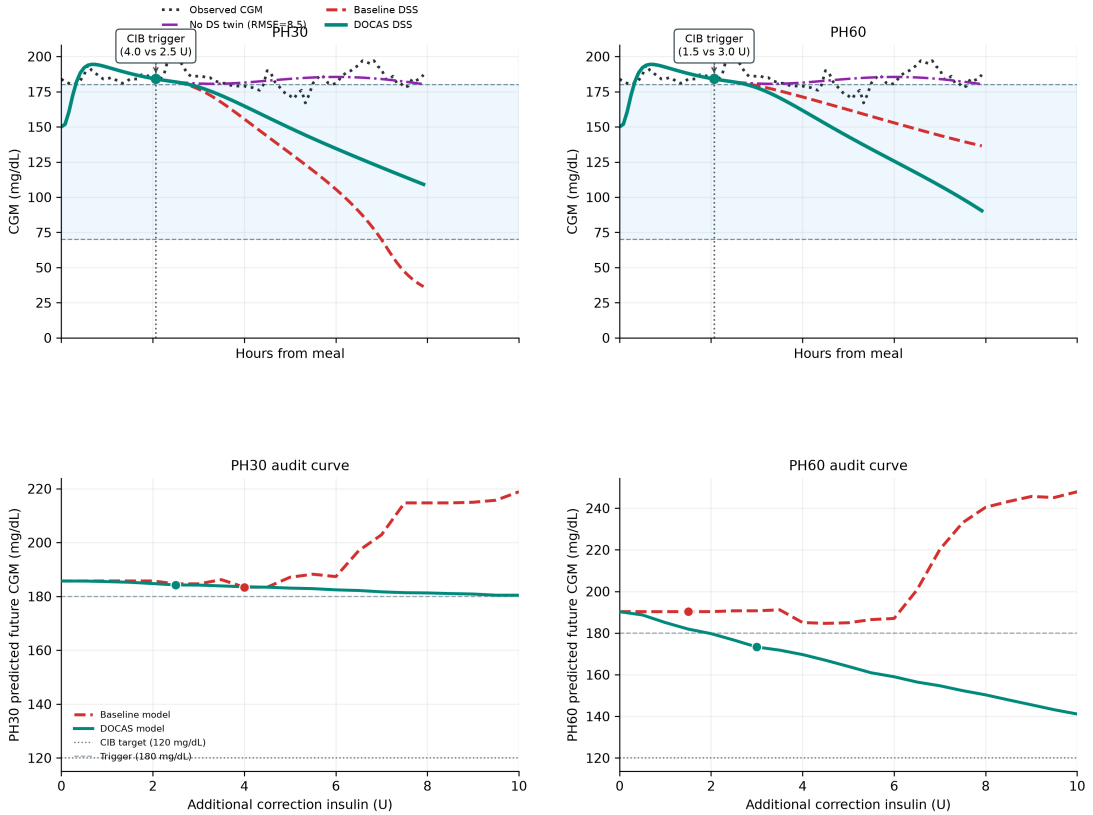


Figure 4: Representative patient-588 ReplayBG window (same postprandial period; twin RMSE 8.5 mg/dL). **Top:** simulated CGM trajectories for the observed data, the untreated digital twin (No DS), and the baseline and DOCAS decision-support controllers at PH30 (left) and PH60 (right); shaded band marks the 70–180 mg/dL target range; dotted vertical line marks the corrective-bolus trigger (4.0 vs. 2.5 U at PH30; 1.5 vs. 3.0 U at PH60). **Bottom:** interventional audit curves at the first CIB decision context for each horizon (markers: delivered bolus). The baseline audit curve predicts higher future glucose with more insulin; the DOCAS audit curve predicts lowering at both horizons, explaining the safer PH30 bolus and the stronger PH60 correction in the upper panels.

flexibility with five ablations on patient 588 at PH30 (Table 4, Figure 5), each starting from the same baseline model and deliberately using simple non-ReplayBG targets.

Hardcoded hyperbolic target. Forcing a fixed hyperbolic target (no ReplayBG kinetics) drives alignment error to 2.9 mg/dL—comparable to the main DOCAS model’s 3.8 mg/dL on the same participant—but at an RMSE cost of +5.4 mg/dL, whereas ReplayBG-preset DOCAS essentially preserves RMSE on the cohort (Figure 5, panel 1; Table 2). A physiologically scaled τ therefore buys most of the alignment benefit without the accuracy cost of an aggressive

synthetic amplitude; brute-force fitting can always buy more alignment, but the price is a model that no longer tracks real glucose dynamics away from the synthetic grid.

Feature independence. Setting the target amplitude to zero forces a perfectly flat target function: insulin is given *no* effect on the forecast whatsoever (panel 2). The fitted audit curve reproduces this null shape almost exactly (alignment error 1.5 mg/dL) at a small RMSE cost (+1.0 mg/dL), confirming that DOCAS can impose feature independence, not only a specific monotone shape, when that is the desired constraint.

Insulin–CGM interaction (indirect effect). Rather than shifting the forecast directly, we let insulin modulate *how* strongly the current CGM value drives the forecast: the target sets $\partial \hat{g} / \partial \text{CGM} = 1$ at zero insulin (full pass-through) and $1 - \delta_{30} = 0.85$ at the maximum corrective dose. The fitted model recovers this slope schedule almost exactly ($1.00 \rightarrow 0.86$ learned vs. $1.00 \rightarrow 0.85$ target; panel 3), showing that the same mechanism can encode indirect, interaction-type physiological priors—insulin’s effect need not act on the forecast in isolation, but can act on another feature’s effect.

Two-dimensional insulin–CHO surface. Finally we combine a hyperbolic insulin-lowering effect with a linear carbohydrate-driven rise in a single target surface, $\tau(x, u, c) = f_0(x)(1 + \delta(c_{\text{norm}} - r(u)))$ (panel 4). The fitted surface tracks the target closely (surface RMSE 4.8 mg/dL) at a modest RMSE cost (+1.2 mg/dL), demonstrating that target-curve alignment extends beyond one-dimensional insulin curves to joint multi-feature target surfaces.

Full glycaemic range. Restricting synthetic anchors to hyperglycaemia ($f_0 > 170$ mg/dL) is intentional: the same hyperbolic τ applied across the full band down to the time-in-range floor ($f_0 > 70$ mg/dL) reaches strong full-band alignment (2.5 mg/dL) but at an RMSE cost of +5.8 mg/dL—on par with aggressive hyper-only hardcoded fitting and far above the near-zero ReplayBG-preset DOCAS RMSE delta (panel 5). Extending alignment into normoglycaemia therefore buys little clinically for corrective-bolus DSS (which triggers only above 180 mg/dL) while further eroding observational forecasting skill; the hyperglycaemia-only context rule is what keeps the RMSE–alignment trade-off manageable.

5. Discussion

This study demonstrates that observational glucose models can be accurate yet *interventionally wrong*: they fit CGM trajectories while learning associative patterns induced by covariate bias between insulin, meals, and glycaemic state—high insulin predicts high future glucose because boluses are delivered into hyperglycaemic, meal-overlapping contexts [7, 13]. SHAP and related explainability tools—including causal, do-operator attributions such as do-SHAP [30, 31]—are necessary to *diagnose* such failures but, by themselves, only *measure* the model and do not repair the learned interventional mapping [8, 15]. DOCAS complements explanation with *alignment*: it matches the model’s interventional audit curve to a stated target function τ . Synthetic rows sampled along τ teach the audit curve how future glucose should change when insulin is varied holding other features fixed—the counterfactual probe $do(\text{insulin} = u)$ that a bolus-sizing DSS actually issues [7, 12, 22].

The target in Eqs. (1)–(2) encodes the same delayed subcutaneous absorption, remote insulin action, glucose kinetics, and interstitial lag used in the downstream ReplayBG evaluation [12]. This matters qualitatively: ReplayBG predicts little correction-bolus effect by 30 minutes but a much larger effect by 60 minutes, rather than a dose curve that falls most steeply immediately and then saturates. By restricting supervision to hyperglycaemic contexts above 170 mg/dL, DOCAS focuses alignment where corrective semantics apply and avoids imposing large artificial drops on normoglycaemic states. The resulting curves preserve the monotonicity properties Kushner et al. verify formally in neural predictors [13] while retaining simulator-derived horizon and dose dependence.

Table 2 makes these contrasts quantitative on the same Ohio protocol. Hard monotone LightGBM constraints reduce PH30 alignment error only from 14.3 to 10.7 mg/dL; p-LSTM’s physiological IOB/COB filters reach 7.7 mg/dL; DOCAS LightGBM and DOCAS LSTM reach 3.5 and 1.9 mg/dL respectively, with a similar ordering at PH60. Relative to architectural approaches such as the physiological preprocessing layer in p-LSTM [7, 18, 19], DOCAS is model-agnostic and optimizes the interventional audit curve directly without redesigning the network—and still improves the already-filtered p-LSTM backbone. Relative to monotonicity constraints, mechanistic LightGBM feature engineering [38], or conformance penalties [13, 23], it accepts arbitrary target curves—not only directional signs—and separates *where* to align (context rule) from *what* to align (τ). Relative to full digital-twin or PINN hybrids [12, 46], it requires no online identification of differential-equation parameters. Packaging DOCAS as an open-source library makes this flexibility explicit: practitioners in other domains can plug in their own target and context logic while reusing the same augmentation loop.

Table 4

PH30 ablations on patient 588: alternative DOCAS targets. Align. (mg/dL RMSE to τ ; full-range uses $f_0 > 70$ mg/dL, others use hyperglycaemic contexts) and RMSE Δ vs. baseline (20.69 mg/dL).

Ablation	Align.	RMSE Δ
Hardcoded (hyper only)	2.89	+5.36
Independence	1.49	+0.95
CGM interaction	0.01	+2.04
CHO surface	4.79	+1.24
Full range ($f_0 > 70$)	2.47	+5.76

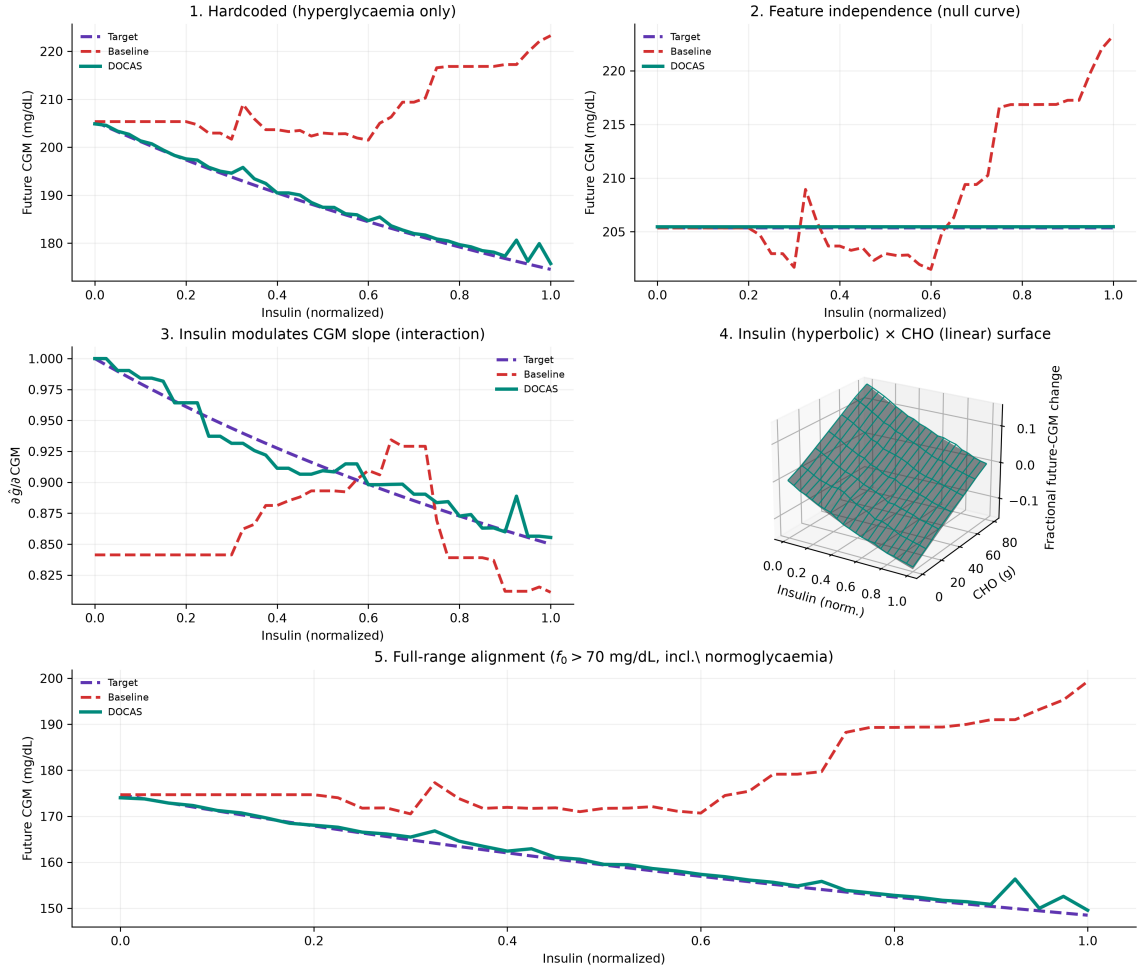


Figure 5: Ablation study, patient 588, PH30. **1:** hardcoded, untuned augmentation on hyperglycaemic anchors reaches tight alignment at a high RMSE cost. **2:** a zero-amplitude target forces feature independence—a flat target function. **3:** an insulin–CGM interaction target changes the slope of CGM’s effect on the forecast (indirect effect) rather than shifting the forecast directly. **4:** a two-dimensional target surface combines hyperbolic insulin lowering with a linear CHO-driven rise; the learned audit surface (wireframe) tracks the target surface (shaded). **5:** the same hyperbolic τ as in panel 1, but with anchors spanning the full glycaemic band ($f_0 > 70$ mg/dL), including normoglycaemia.

Beyond T1D, the DOCAS software interface formalizes a general recipe for injecting prior knowledge into black-box models under covariate bias: (1) specify the intervention feature; (2) define a context rule and target function τ from domain theory; (3) augment observational training data with synthetic (x, a, τ) tuples sampled on selected

anchors; and (4) validate audit-curve alignment on held-out counterfactual probes, not only on observational error. Because model, context, and target are all user-supplied, the same code path covers pharmacological dose-finding with limited interventional data [11], critical-care vasopressor titration, and settings where scanner metadata confounds predictions [47, 48]—each with a different τ and anchor predicate. Process-driven synthetic data are already accepted in drug development when they preserve decision-relevant inferences [44]; DOCAS applies the same philosophy at the level of *model behaviour* rather than dataset generation.

The cost of alignment is explicit in our T1D experiments, but with the ReplayBG preset it is small: cohort-mean RMSE is essentially unchanged when alignment is imposed—a compromise in the same spirit as the small accuracy gap between p-LSTM and np-LSTM in Prendin et al., which nonetheless produced large DSS differences [7]—but that cost is not uniform under more aggressive targets (Section 4.3). Per-participant RMSE deltas and alignment gains vary widely (Figure 3), and it remains unclear why alignment is comparatively cheap for some patients yet demands a larger forecasting penalty for others. Likely drivers include how strongly insulin is confounded with hyperglycaemia in each subject, how much observational trajectories already resemble the imposed target, and broader cohort heterogeneity in dosing behaviour and glycaemic variability; we do not yet know which of these dominates. For decision support where a single erroneous bolus can cause hypoglycaemia, a bounded trade-off is clinically acceptable when the alignment benefit is real [45, 49], but understanding when that compromise is unavoidable remains an open question.

Limitations. In the present T1D preset, alignment is deliberately *partial*: synthetic anchors and the audit probe are restricted to hyperglycaemic contexts ($f_0 > 170$ mg/dL), so the interventional curve is not constrained under normoglycaemia or mild elevation. Extending τ across the full band down to the time-in-range floor ($f_0 > 70$ mg/dL) improves full-band alignment but inflates held-out RMSE by $\approx +6$ mg/dL on patient 588 (Section 4.3), versus a near-zero RMSE delta for hyperglycaemia-only ReplayBG-preset DOCAS—confirming that broader context coverage competes directly with forecasting skill where a corrective-bolus DSS does not need it. The current target uses ReplayBG prior-mode parameters and patient body weight, but not patient-specific twinned insulin sensitivity, meal state, or the age distribution implicit in the IOB feature. It therefore represents a newly delivered correction as a separate added-dose channel alongside observed IOB, matching the replay controller probe but remaining a reproducible population prior rather than a fully individualized digital twin. Future work could condition the target on prospectively identified ReplayBG parameters and latent state [12, 50]. ReplayBG evaluation is currently restricted to patient 588; extending to additional OhioT1DM subjects and to prospective closed-loop trials remains necessary. DOCAS does not replace causal identification when unobserved confounders dominate—SHAP attributions can still mislead on individual instances, as Prendin et al. note for exercise and illness [7]. Finally, synthetic augmentation assumes the target function τ is correct; misspecified priors could imprint a wrong interventional response as readily as observational bias imprints a wrong associational one.

Future directions. A first priority is to narrow any residual RMSE–alignment trade-off under aggressive or misspecified targets. Our T1D ReplayBG preset already obtains tight audit-curve alignment at negligible forecasting cost; it remains open whether richer target families, multi-horizon targets, or staged fitting that separates alignment from accuracy tuning can keep that property when τ is less physiologically matched. Closely related, we need to explain the large per-patient spread in that compromise: why alignment is near-costless for some OhioT1DM participants but expensive for others, and how subject-level confounding strength, baseline misalignment, and cohort heterogeneity jointly determine where synthetic augmentation can correct behaviour without eroding forecasting skill. Second, we plan to apply the same synthetic-augmentation interface beyond glucose forecasting to other biomedical datasets where observational training induces covariate bias: treatment timing or dose collinear with severity (intensive care, oncology dosing, pharmacovigilance), behavioural interventions delivered when risk is already elevated (readmission, deterioration alerts), and any setting where a model must answer a counterfactual “what if we changed treatment?” rather than only fit associative patterns. Third, runtime remains a practical barrier for heavier architectures: each residual-correction pass retrains on real plus synthetic rows, which is manageable for gradient-boosted trees but slower for neural nets. Faster variants could cache and warm-start repeated fits or substitute lightweight proxies during development. Finally, prospective validation—embedding DOCAS in live or shadow-mode decision support and extending ReplayBG-style replay beyond patient 588—is needed before clinical deployment.

6. Conclusion

We presented DOCAS, an open-source Python package for synthetic-augmentation alignment that trains a new model on real and synthetic rows toward a user-specified target function τ . In type 1 diabetes we instantiate τ as a ReplayBG-derived correction-bolus response; on OhioT1DM, DOCAS outperforms baseline LightGBM, monotone-constrained LightGBM, and Prendin np-/p-LSTM on held-out audit-curve alignment while essentially preserving forecasting accuracy on both tree and LSTM backbones, and in ReplayBG simulation an aligned LightGBM controller increases time-in-range and reduces hypoglycaemia relative to an equally accurate but misaligned baseline. Because practitioners choose the model, context rule, and target function, DOCAS is more flexible than architecture-locked or monotonicity-only repair strategies. For decision support and hybrid closed-loop systems, validating interventional audit curves against stated target functions should be as routine as reporting RMSE; DOCAS offers a reusable path to that validation wherever domain knowledge specifies how predictions ought to change under intervention.

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